

Observations from A1

No **perfect** visualization.

Different people have different opinions,
but there are consensus.

Less is usually better.

The most popular example

A circular sunburst chart illustrating the global distribution of COVID-19 cases as of April 8, 2020. Each segment's area represents the total number of cases in a specific country or region. The segments are color-coded and labeled with the country name and its corresponding case count.

Country	Cases
India	1,429
China	1,426
USA	340
Mexico	129
Canada	38.8
Brazil	216
Argentina	45.8
Peru	34.4
Colombia	52.1
Venezuela	28.8
Spain	47.5
Germany	83.3
Italy	58.9
France	64.8
UK	67.7
Russia	144
Egypt	113
Turkey	85.8
Yemen	34.4
Iraq	45.5
Uzbekistan	35.2
Afghanistan	42.2
Sri Lanka	21.9
Bangladesh	173
Pakistan	241
Nepal	30.9
Malaysia	34.3
Philippines	117
S. Korea	51.9
Mali	23.3
Sudan	48.1
Tanzania	67.4
Congo DRC	102
Côte d'Ivoire	28.9
Ghana	34.1
Kenya	55.1
Niger	27.2
Ethiopia	127
Mozambique	33.9
Zambia	20.6
Burkina Faso	23.3
Madagascar	30.3
Cameroon	28.6
Nigeria	224
Angola	36.7
Morocco	37.8
Malawi	20.9

Information is Beautiful // United Nations mid-2023 projection // un-ilibrary.org // May 2023 / [data](#)

U.S. GDP by Region

Region	GDP
Far West	\$5.8T
Southwest	\$3.7T
South	\$3.2T
Midwest	\$2.8T
Northeast	\$2.5T
Southeast	\$2.2T

State	GDP
California	\$4.1T
Texas	\$2.7T
Illinois	\$1.1T
New York	\$2.3T
Florida	\$1.0T
Massachusetts	\$785B
Ohio	\$933B
Michigan	\$711B
Indiana	\$531B
Washington	\$860B
Arizona	\$556B
N. Carolina	\$845B
Virginia	\$770B
Tennessee	\$553B
Alabama	\$324B
Mississippi	\$159B
Oklahoma	\$267B
Oregon	\$333B
Utah	\$303B
Colorado	\$557B
Connecticut	\$368B
ME	
NH	\$122B
VT	
WY	
MT	
ID	\$130B
Nevada	\$262B
AK	\$116B
HI	\$116B
NM	\$141B
RI	
Wisconsin	\$456B
Maryland	\$546B
Penns.	\$1.0T
WV	\$108B

Overseas Activity* \$160B



Around November 2022, the world will reach a pivotal milestone—8 billion global population. What is the distribution of this population, by region and country?

The infographic illustrates the global population distribution in 2022, totaling 8 billion people. It is divided into seven main regions, each represented by a different color in the circular sunburst chart and the circular pie chart.

World Population Distribution by Region (2022):

Region	Population (Millions)
Asia	4,681M
Europe	747M
North America	506M
South America	439M
Africa	1,207M
Oceania	44M
Rest of World	107M

Asia Population Distribution by Country (2022):

Country	Population (Millions)
China	1,451M
India	1,411M
Japan	126M
Indonesia	280M
Philippines	113M
Vietnam	98M
Myanmar	55M
Thailand	70M
Uzbekistan	35M
Rest of Asia	36M

Europe Population Distribution by Country (2022):

Country	Population (Millions)
Russia	146M
Germany	84M
France	66M
Italy	60M
Spain	47M
UK	69M
Rest of Europe	63M

North America Population Distribution by Country (2022):

Country	Population (Millions)
U.S.	335M
Mexico	132M
Canada	39M
Rest of North America	11M

South America Population Distribution by Country (2022):

Country	Population (Millions)
Brazil	216M
Argentina	46M
Venezuela	28M
Peru	34M
Colombia	52M
Rest of South America	62M

Africa Population Distribution by Country (2022):

Country	Population (Millions)
Nigeria	218M
Ethiopia	122M
Rest of Africa	307M
South Africa	61M
Kenya	57M
Tanzania	64M
Uganda	49M
Angola	35M
MDG	29M
D.R.C.	96M
Egypt	107M
Sudan	46M
Marocco	38M
Other Africa	18M

Oceania Population Distribution by Country (2022):

Country	Population (Millions)
Australia	26M
Other Oceania	18M

Due to standard instances of rounding, totals may not add up perfectly to regional sums

Source: UN Population Division, 2022.

 /visualcapitalist

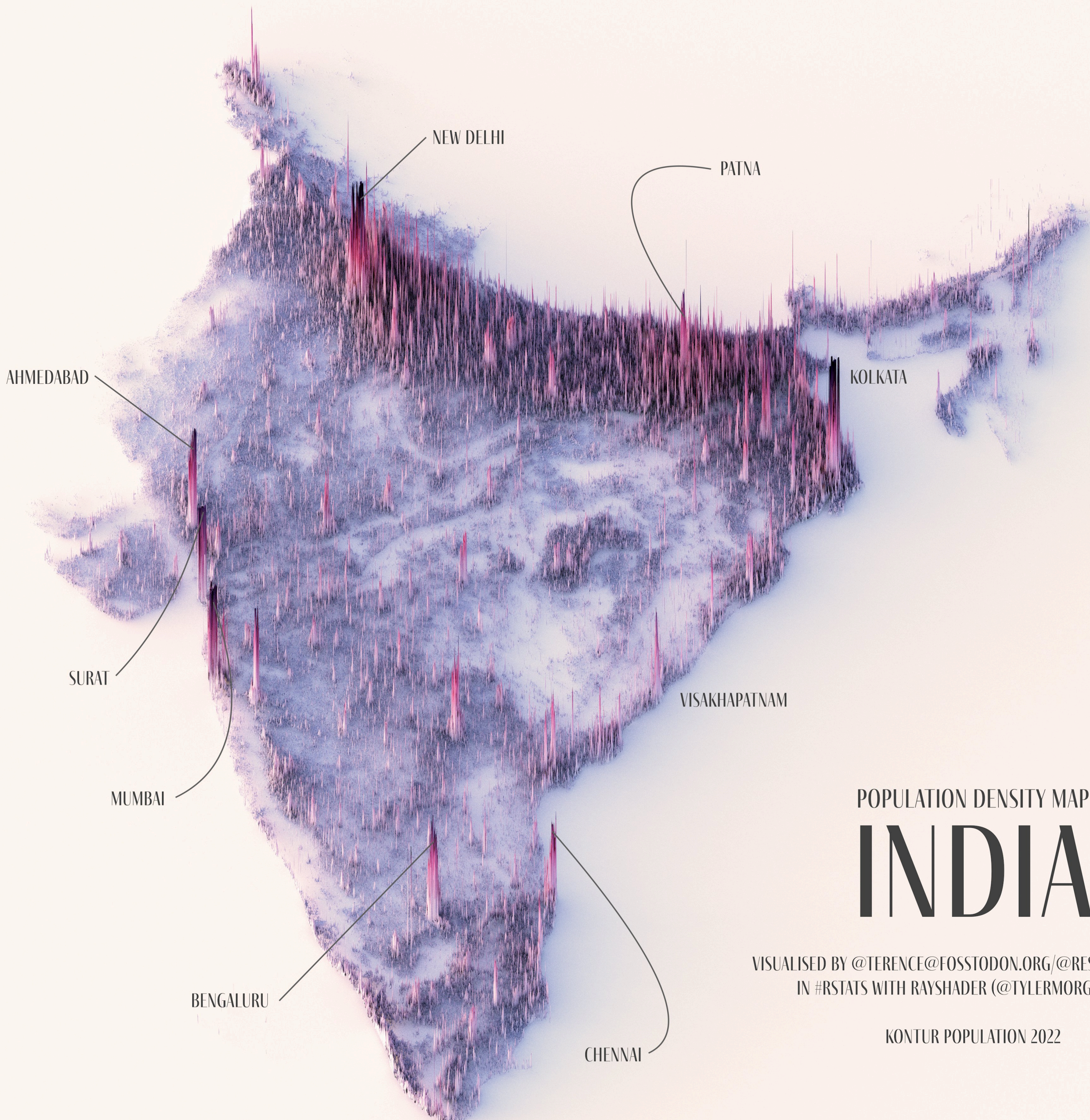


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Well-designed infographics

Hard to make in D3.js but they are beautiful



POPULATION DENSITY MAP

INDIA

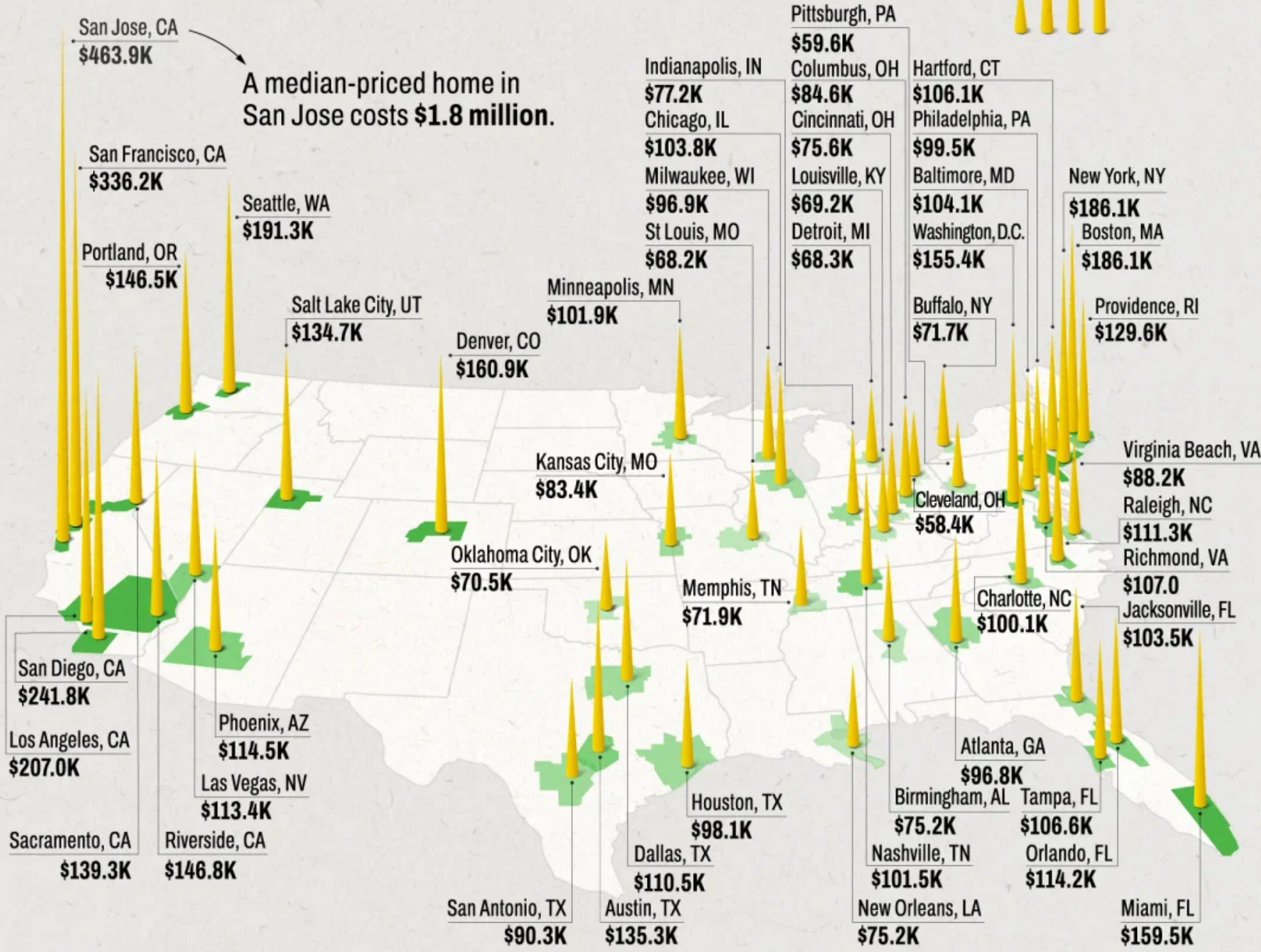
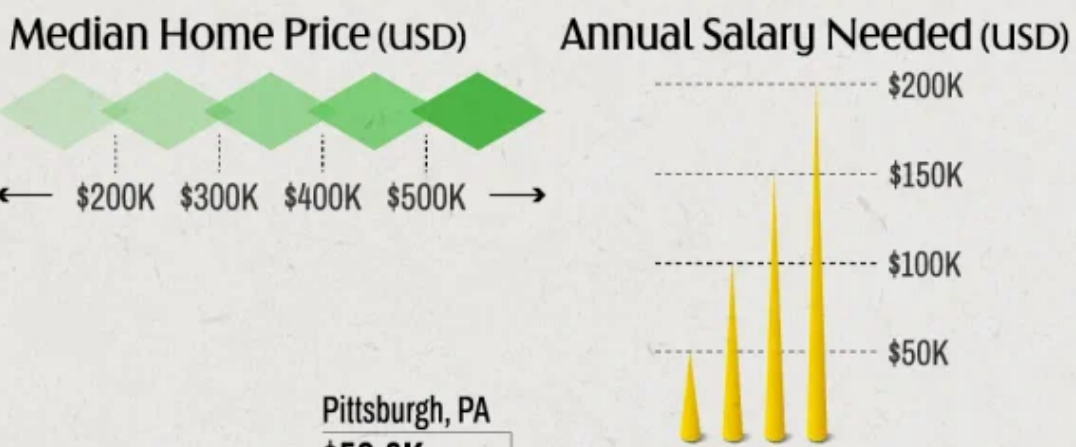
VISUALISED BY @TERENCE@FOSSTODON.ORG/@RESEARCHREMORA
IN #RSTATS WITH RAYSHADER (@TYLERMORGANWALL)

KONTUR POPULATION 2022

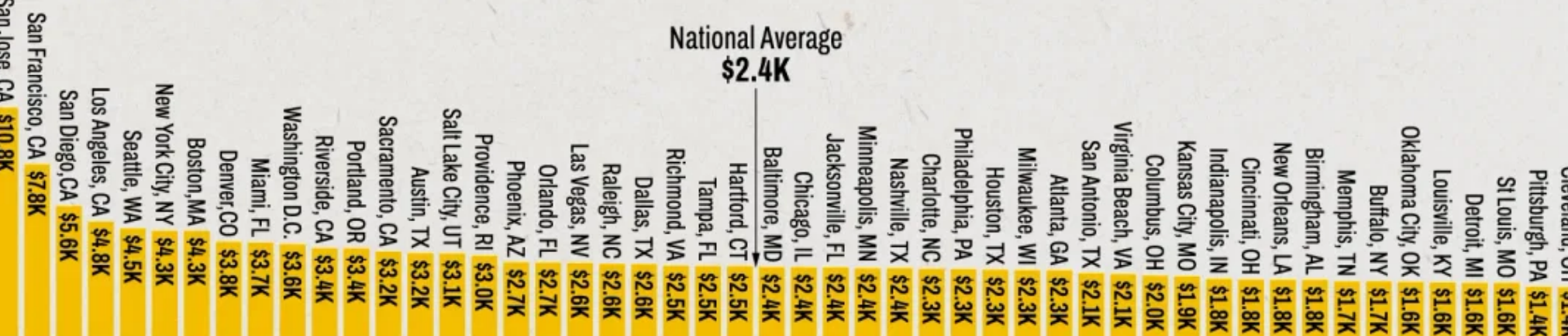
SALARY NEEDED TO BUY A HOME

in 50 U.S. Cities.

The median home price in the U.S. has risen to **\$389,400**, making the salary needed **\$104,339** for a monthly mortgage payment of **\$2,435**.



Ranked: Monthly Mortgage Payment (with taxes and insurance)

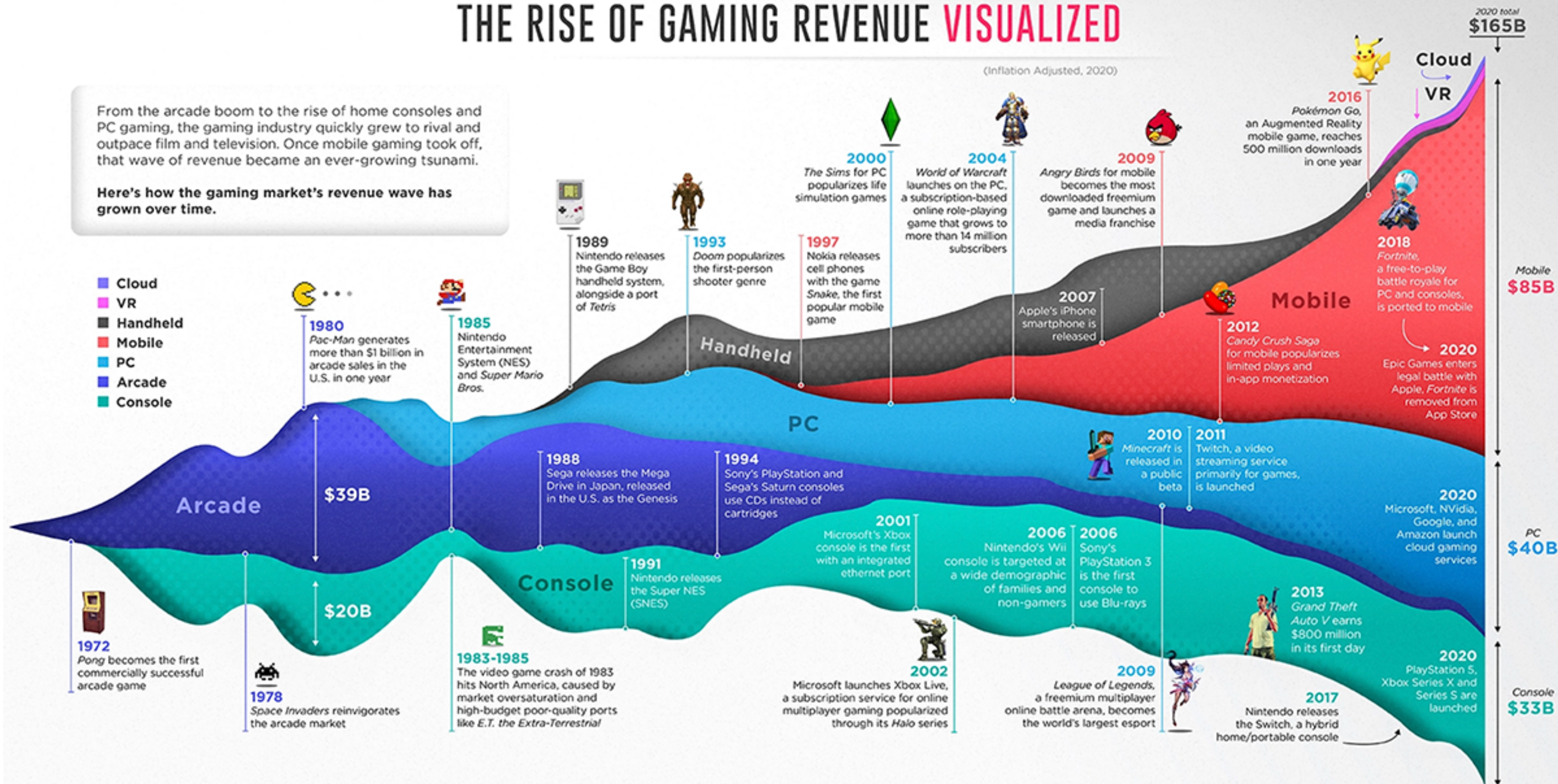


Salary needed to afford the principal, interest, taxes, and insurance payments on a median-priced home in the corresponding metro area.
Source: HSH.com (Data as of May 2024)

THE RISE OF GAMING REVENUE VISUALIZED

From the arcade boom to the rise of home consoles and PC gaming, the gaming industry quickly grew to rival and outpace film and television. Once mobile gaming took off, that wave of revenue became an ever-growing tsunami.

Here's how the gaming market's revenue wave has grown over time.



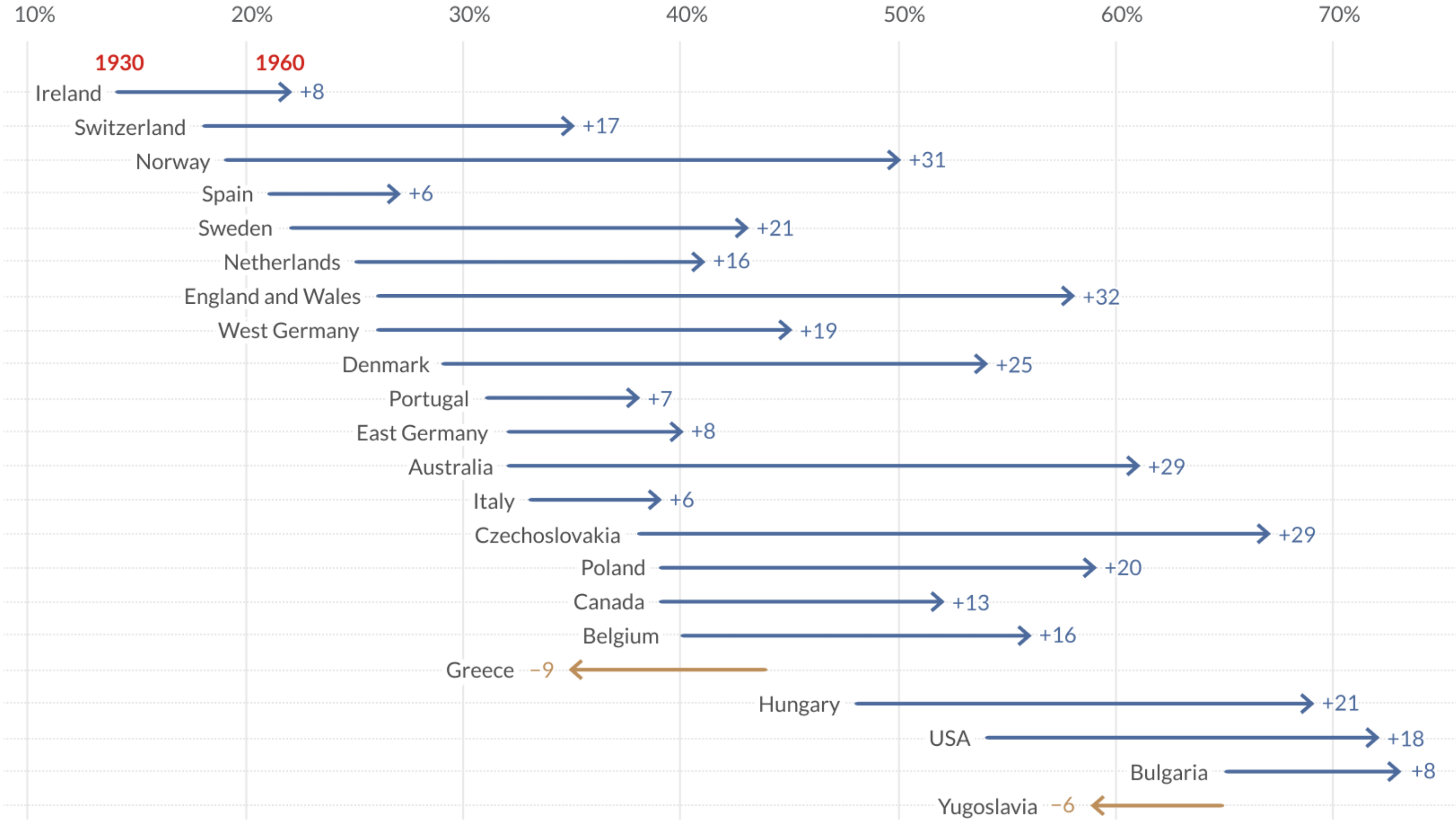
Simple is good

So don't have too much to pick

Marriage rates rose across many high-income countries between 1930 and 1960

Our World
in Data

Share of women ever married, among those aged 20–24

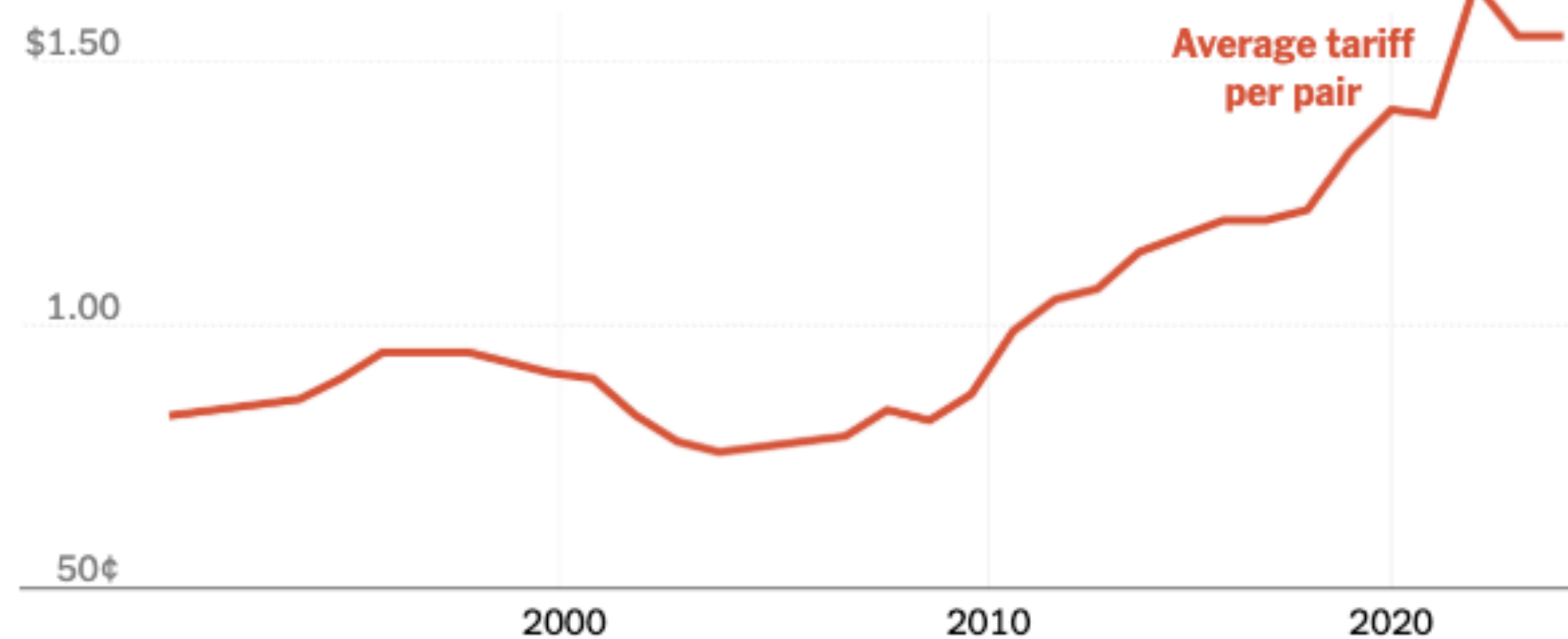


Data source: Jan Van Bavel and David S Reher (2013). *The Baby Boom and Its Causes: What We Know and What We Need to Know*.

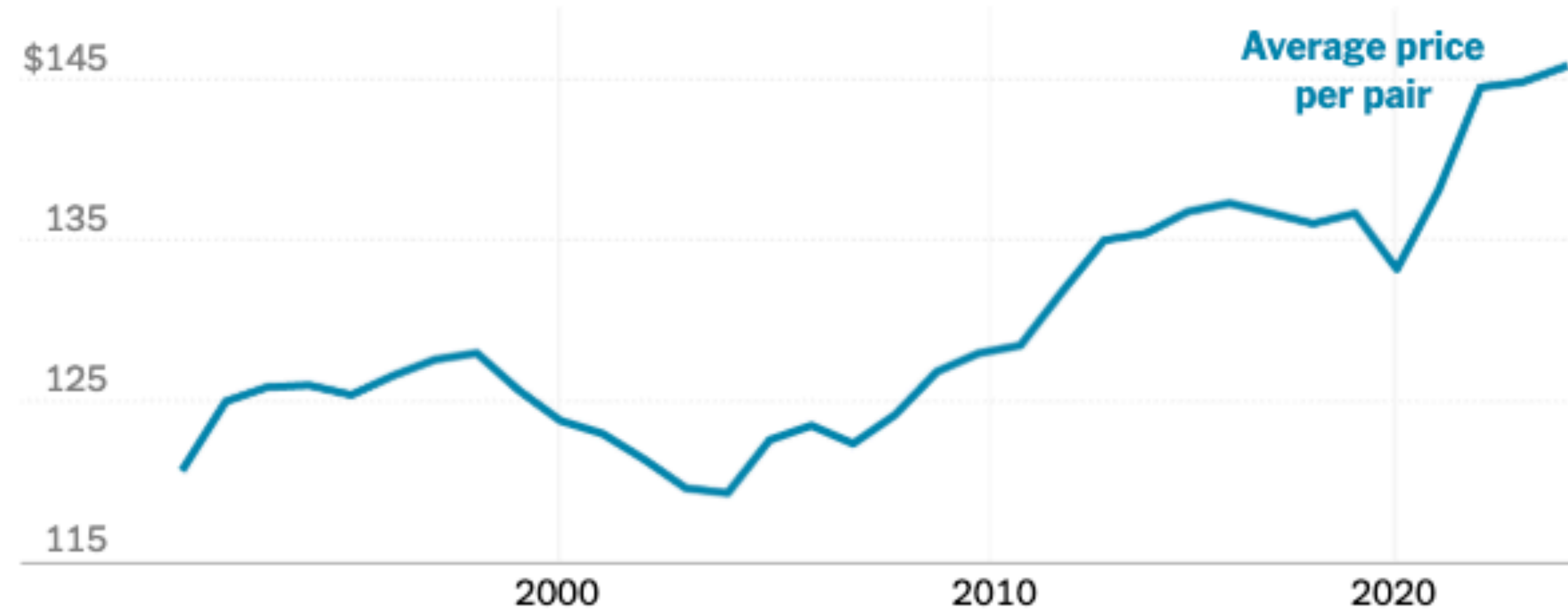
OurWorldinData.org — Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the author Saloni Dattani

Historical data over the past 30 years shows that tariff increases on footwear ...



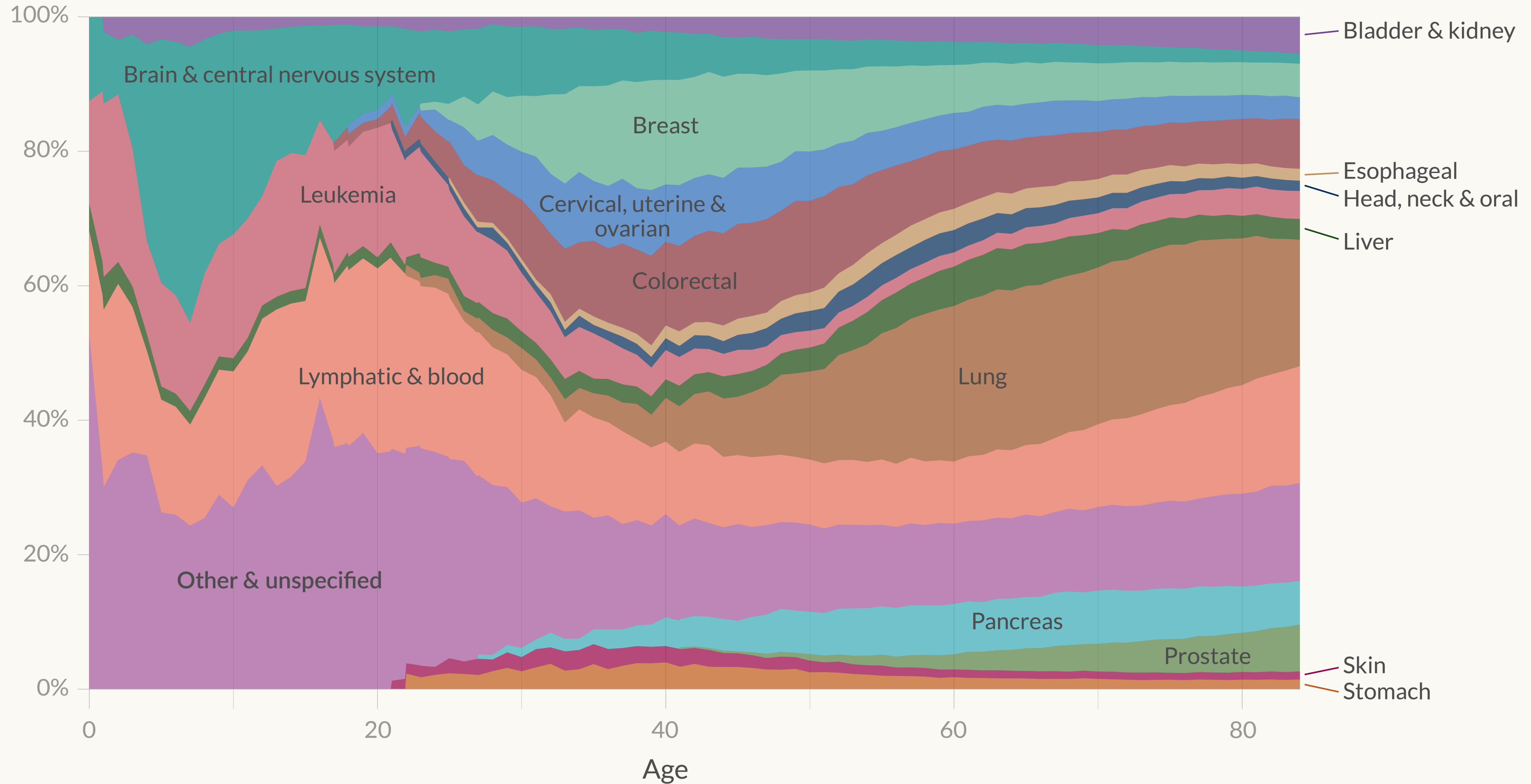
... were consistently followed by higher retail prices



Sources: Data from United States International Trade Commission and Bureau of Labor Statistics. 2024 values are forecasts by Footwear Distributors and Retailers of America based on data through November.

Which cancers do people in the United States die from at different ages?

The **relative share** of cancer deaths from each cancer at each age. Data comes from both sexes between 2018 and 2022. Note that the cancer death rate rises greatly with age, and cancers at a young age are vastly rarer than at older ages.

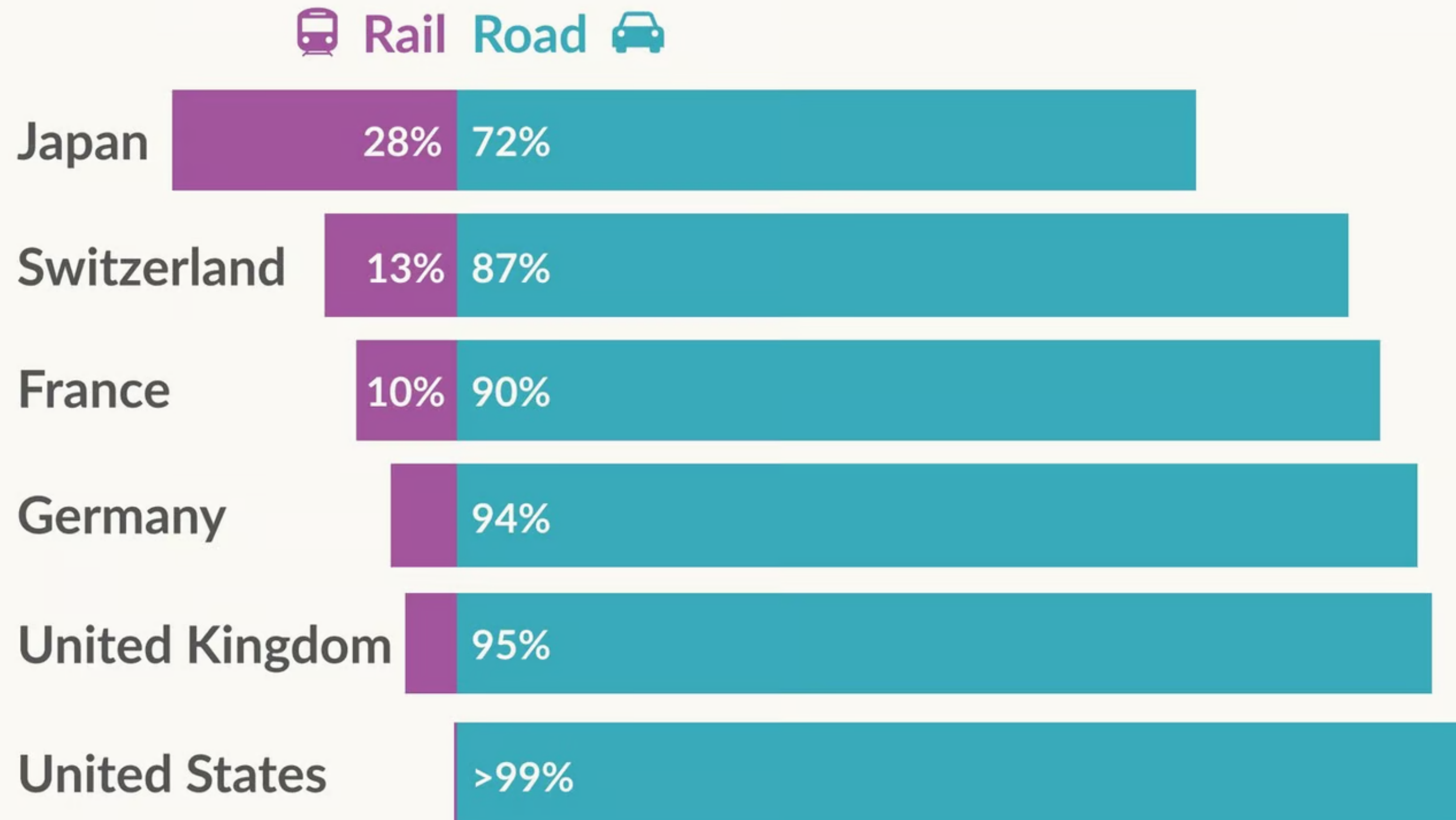


Note: For clarity, several cancer categories have been merged. 'Lymphatic and blood cancers' includes Hodgkin disease, Non-Hodgkin lymphoma, multiple myeloma, and other lymphoid and blood cancers. 'Head, neck, and oral cancers' includes oral, pharyngeal, and laryngeal cancers. 'Other & unspecified cancers' combines in situ, uncertain, benign, and other unspecified cancers.

Source: National data from the 'underlying cause of death' in death certificates, classified with the ICD-10 113 cause list. United States Centers for Disease Control and Prevention, CDC Wonder database (2024)

Rail travel is more common in Japan than in Western Europe and the US

Share of passenger-kilometers — the transport of one passenger over a distance of one kilometer — traveled by rail and road in 2021.



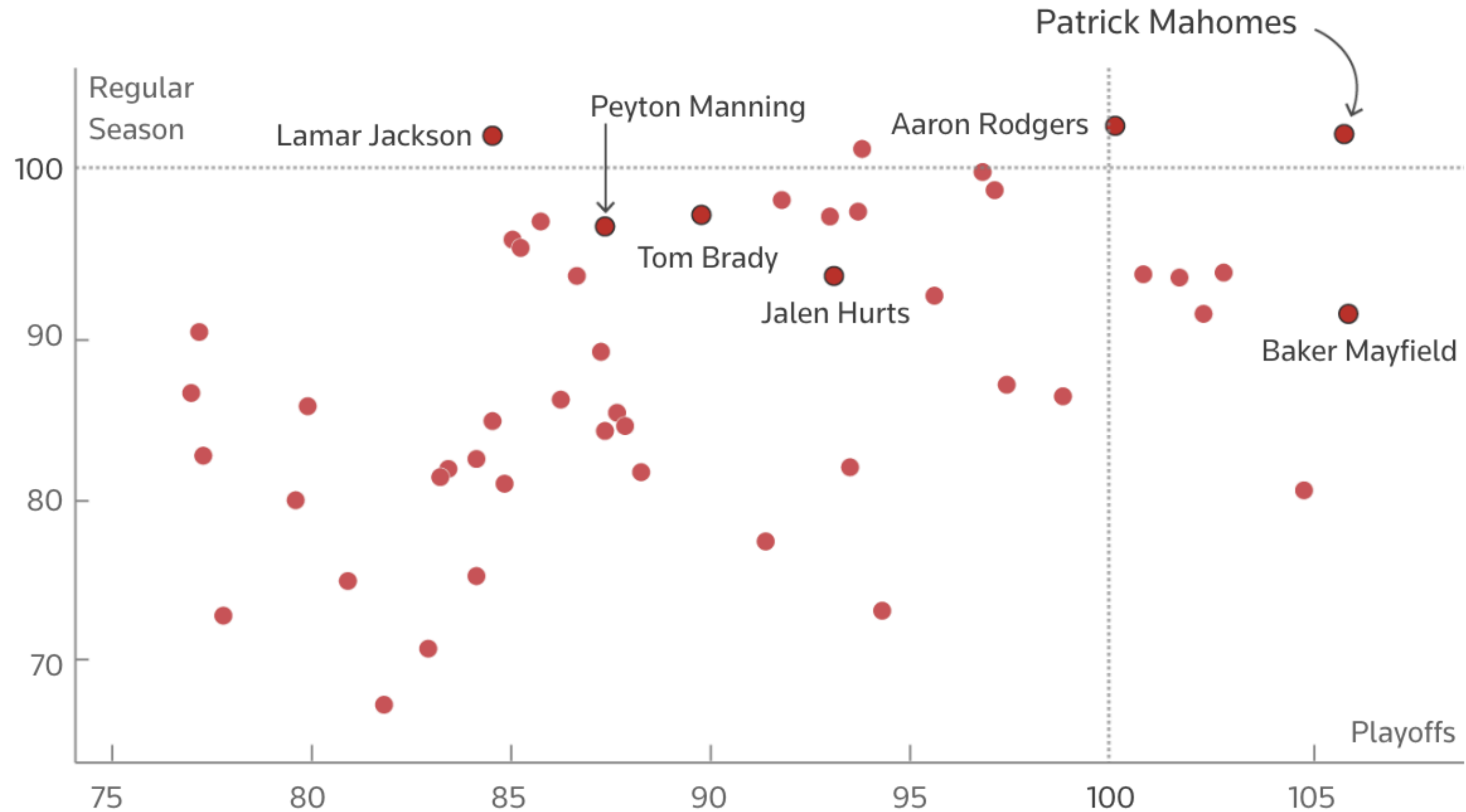
Data source: OECD (2024)

Note: Crew and service personnel are excluded.

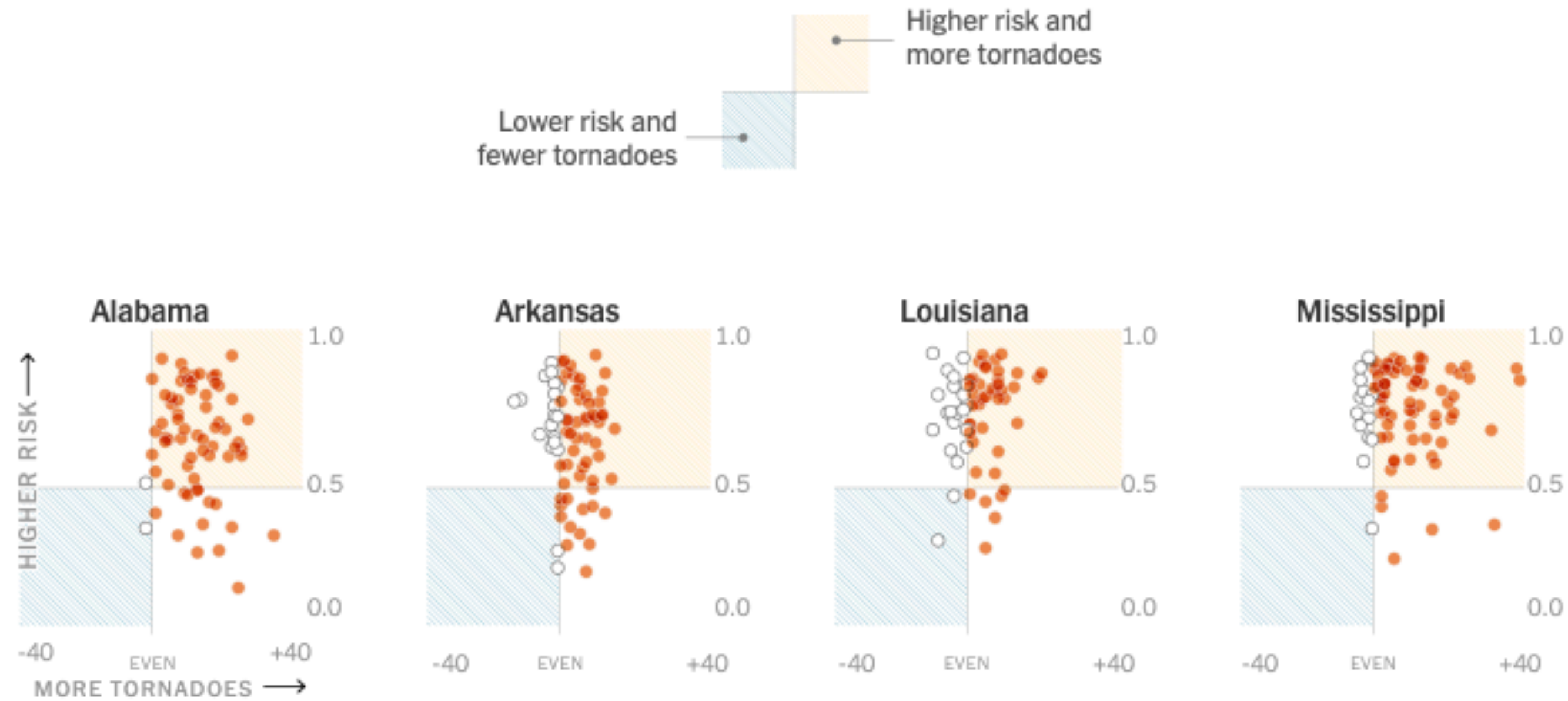
OurWorldInData.org/transport | CC BY

Passer ratings

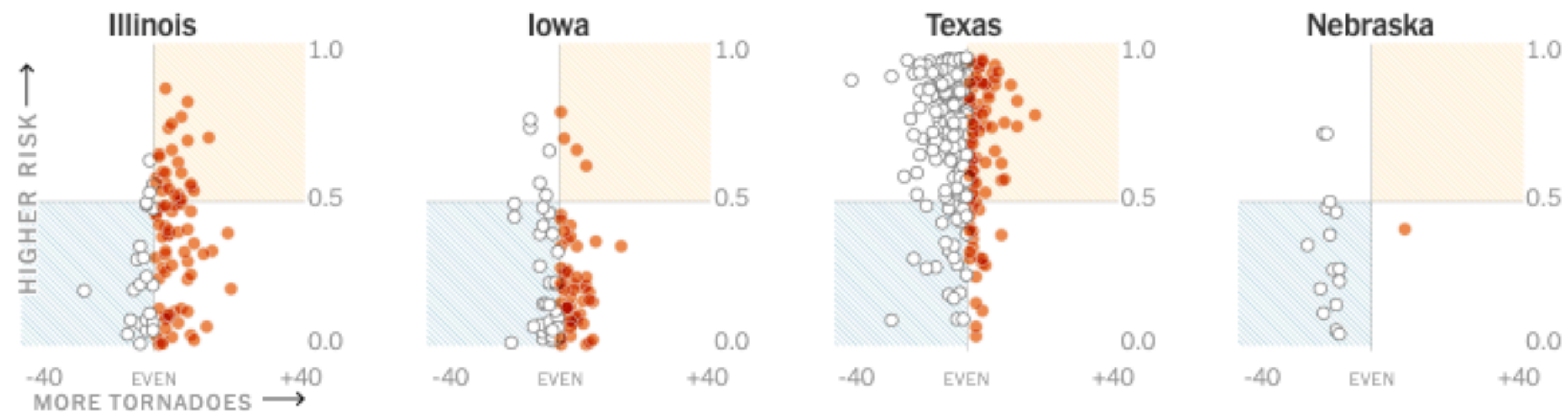
Career ratings for playoffs versus regular season since 1966.



County risk vs. change in tornado activity

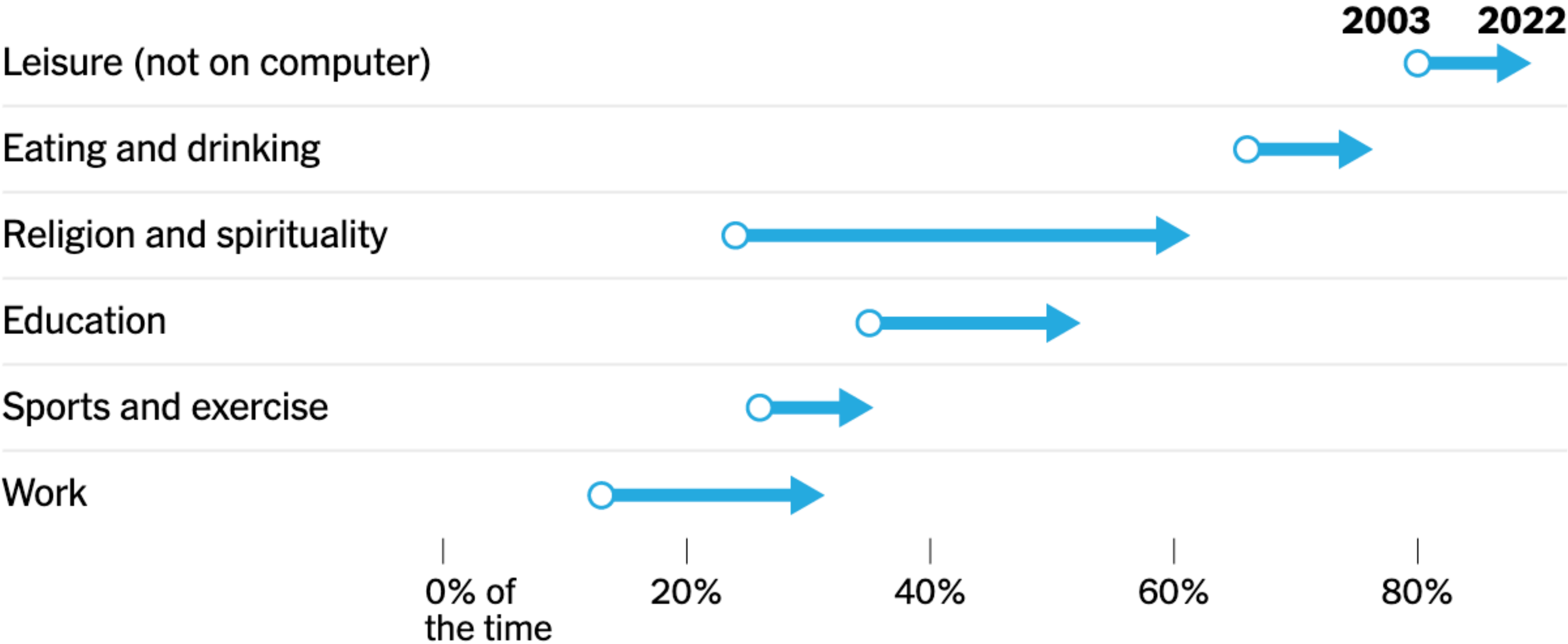


In the states with the most tornadoes this year, most counties have better prepared infrastructure for these kinds of events.



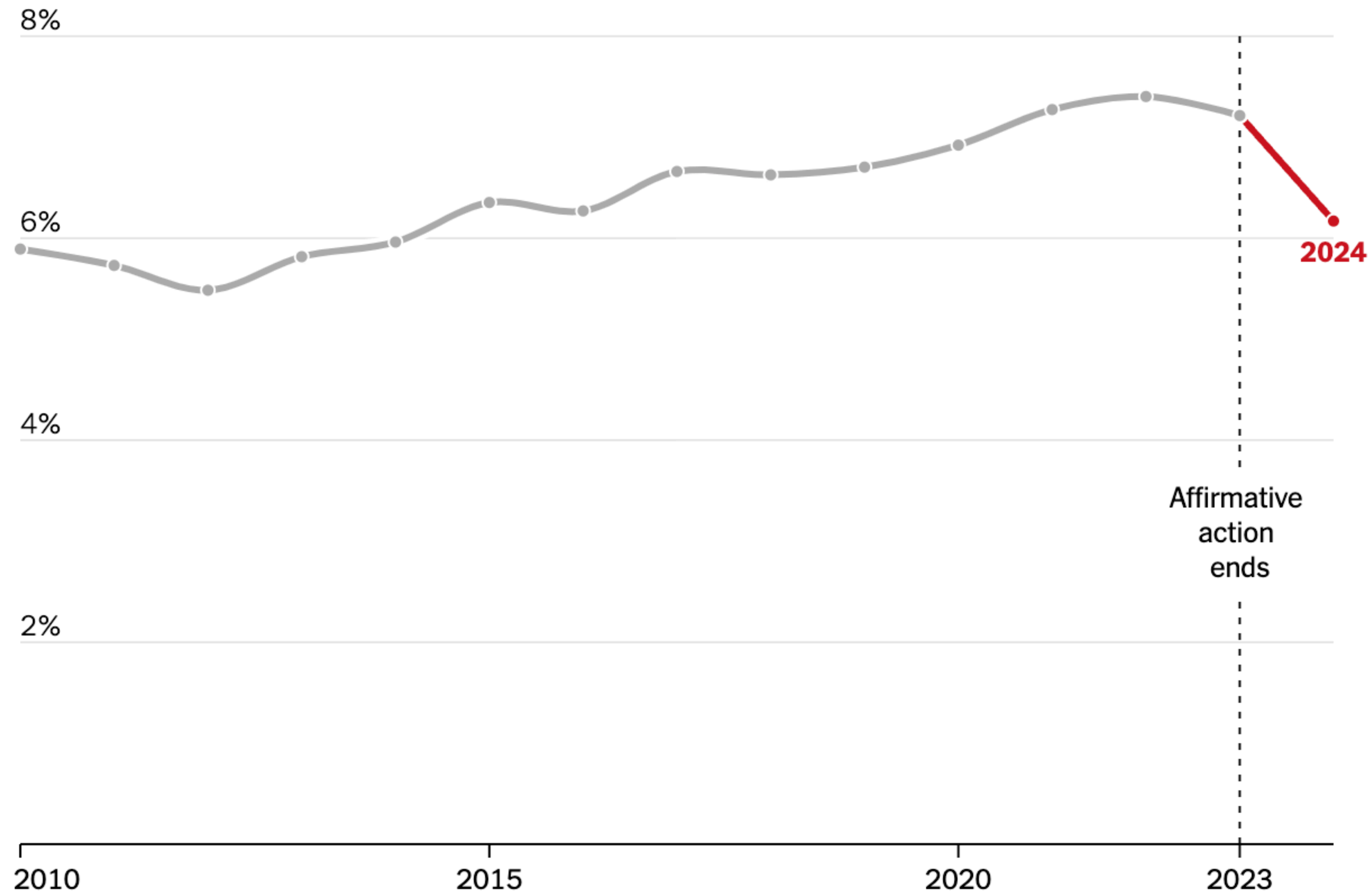
Source: C.D.C. and NOAA • Note: Change in tornado activity compares tornado counts from 2002-22 with 1981-2001.

How often each activity occurred at home



Average share of Black students

Among domestic first-year students at 59 colleges

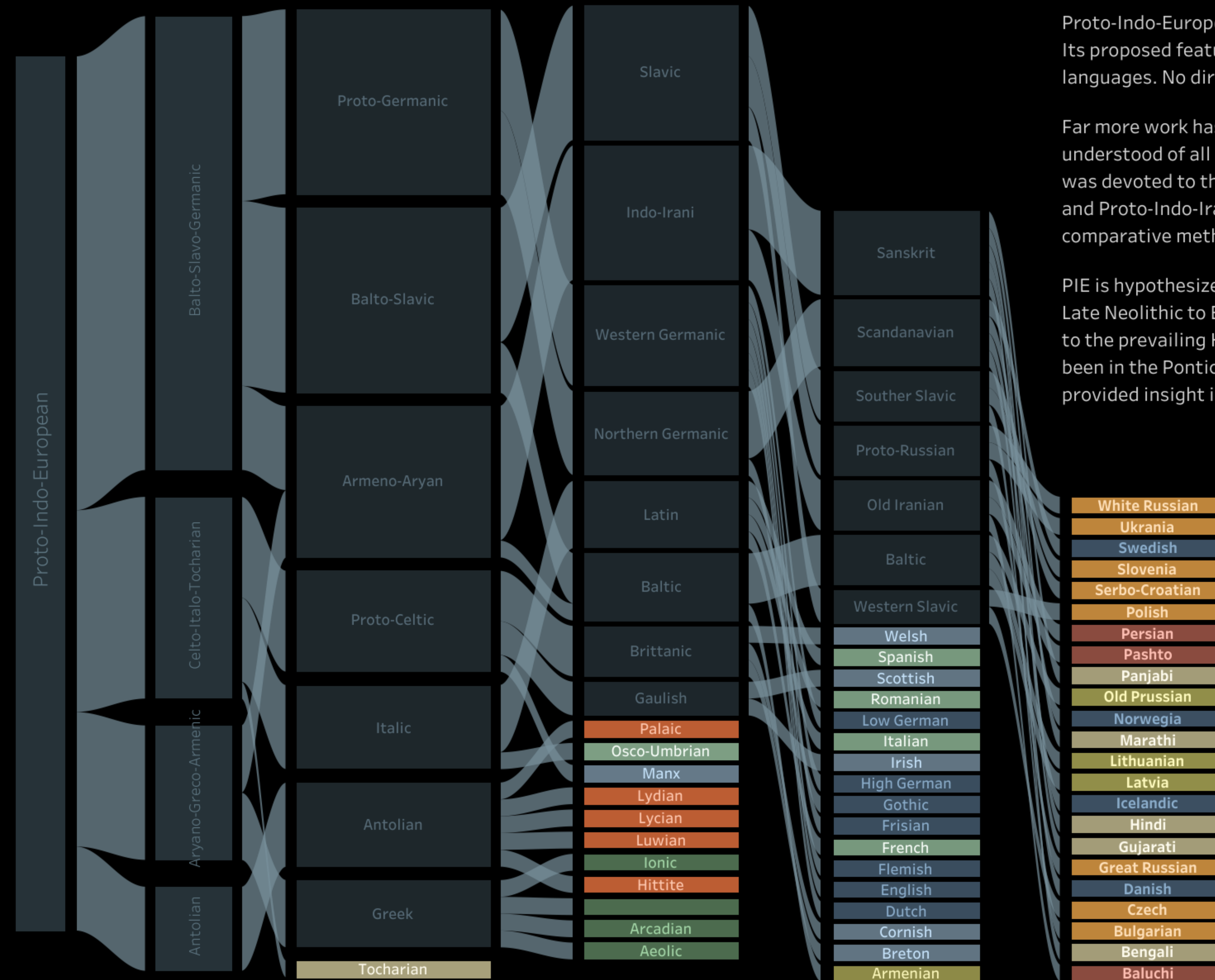


Source: 59 colleges that reported data in a format used by the government (2023–24); [National Center for Education Statistics](#) (2010–22) • Note: The share represents Black students who are not Hispanic or multiracial, following [federal reporting categories](#).

Balanced visualizations

Can be done in D3.js but need some work

THE DEVELOPMENT OF LANGUAGE



Proto-Indo-European (PIE) is the theorized common ancestor of the Indo-European language family. Its proposed features have been derived by linguistic reconstruction from documented Indo-European languages. No direct record of Proto-Indo-European exists.

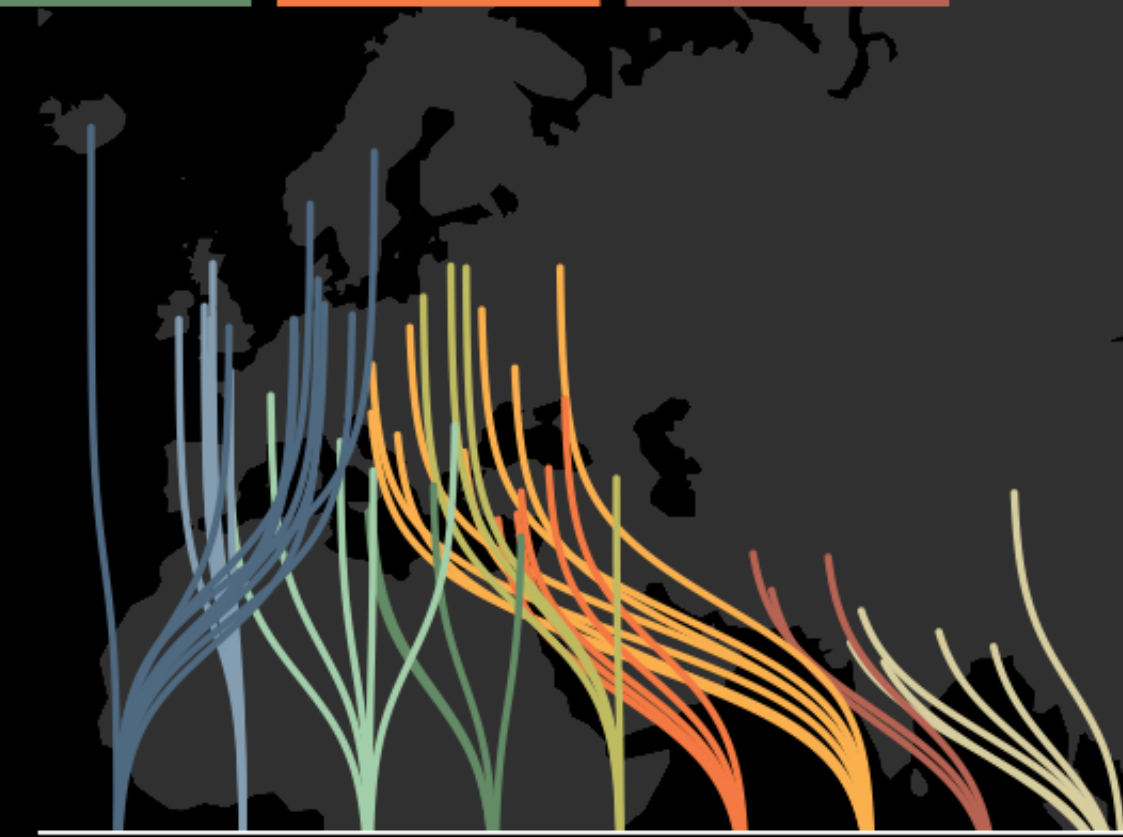
Far more work has gone into reconstructing PIE than any other proto-language, and it is the best understood of all proto-languages of its age. The majority of linguistic work during the 19th century was devoted to the reconstruction of PIE or its daughter proto-languages (such as Proto-Germanic and Proto-Indo-Iranian), and many of the modern techniques of linguistic reconstruction (such as the comparative method) were developed as a result.

PIE is hypothesized to have been spoken as a single language from 4500 BC to 2500 BC during the Late Neolithic to Early Bronze Age, though estimates vary by more than a thousand years. According to the prevailing Kurgan hypothesis, the original homeland of the Proto-Indo-Europeans may have been in the Pontic-Caspian steppe of eastern Europe. The linguistic reconstruction of PIE has provided insight into the pastoral culture and patriarchal religion of its speakers...

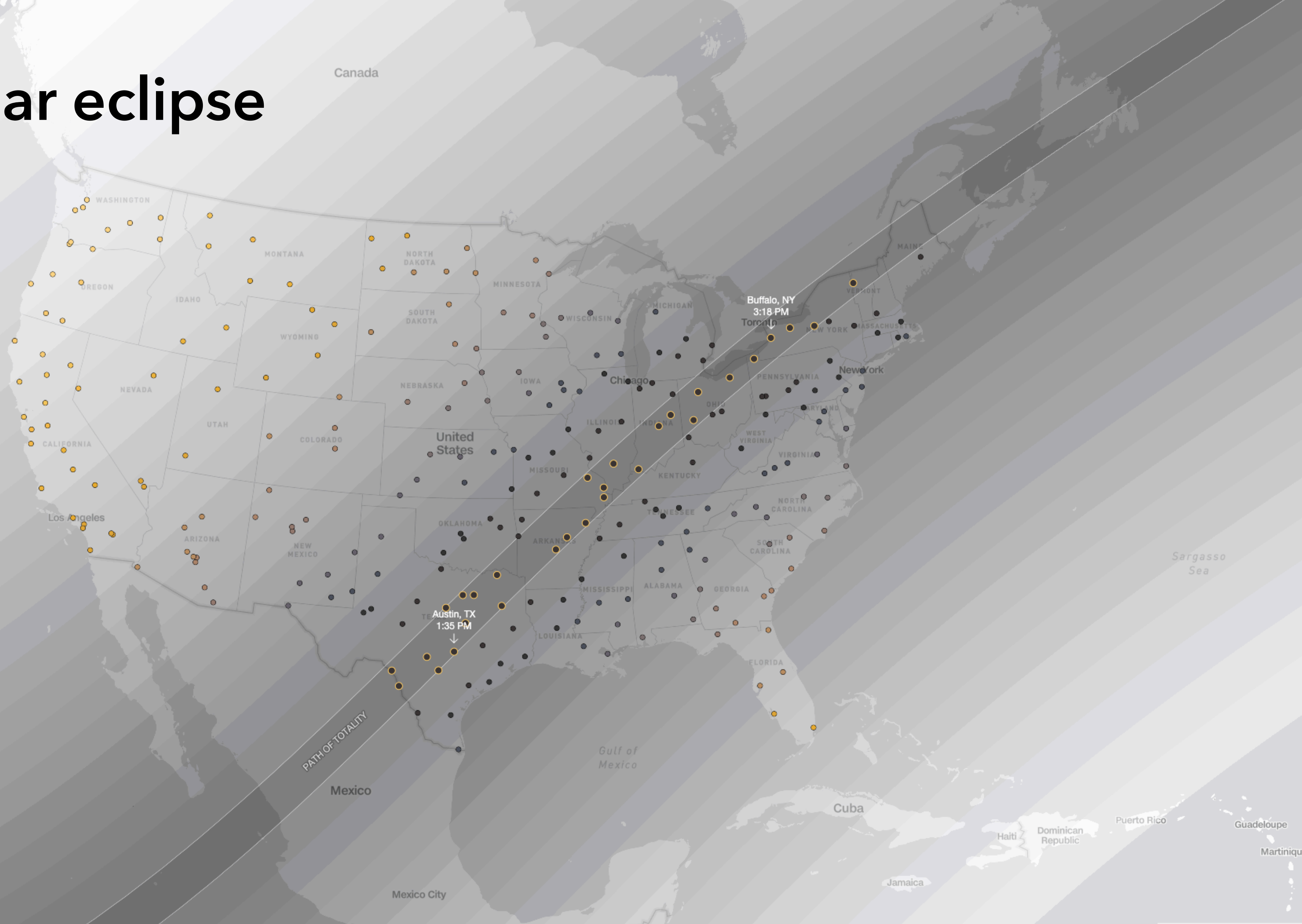
As speakers of Proto-Indo-European became isolated from each other through the Indo-European migrations, the regional dialects of Proto-Indo-European spoken by the various groups diverged, as each dialect underwent shifts in pronunciation (the Indo-European sound laws), morphology, and vocabulary. Over many centuries, these dialects transformed into the known ancient Indo-European languages. From there, further linguistic divergence led to the evolution of their current descendants, the modern Indo-European languages.



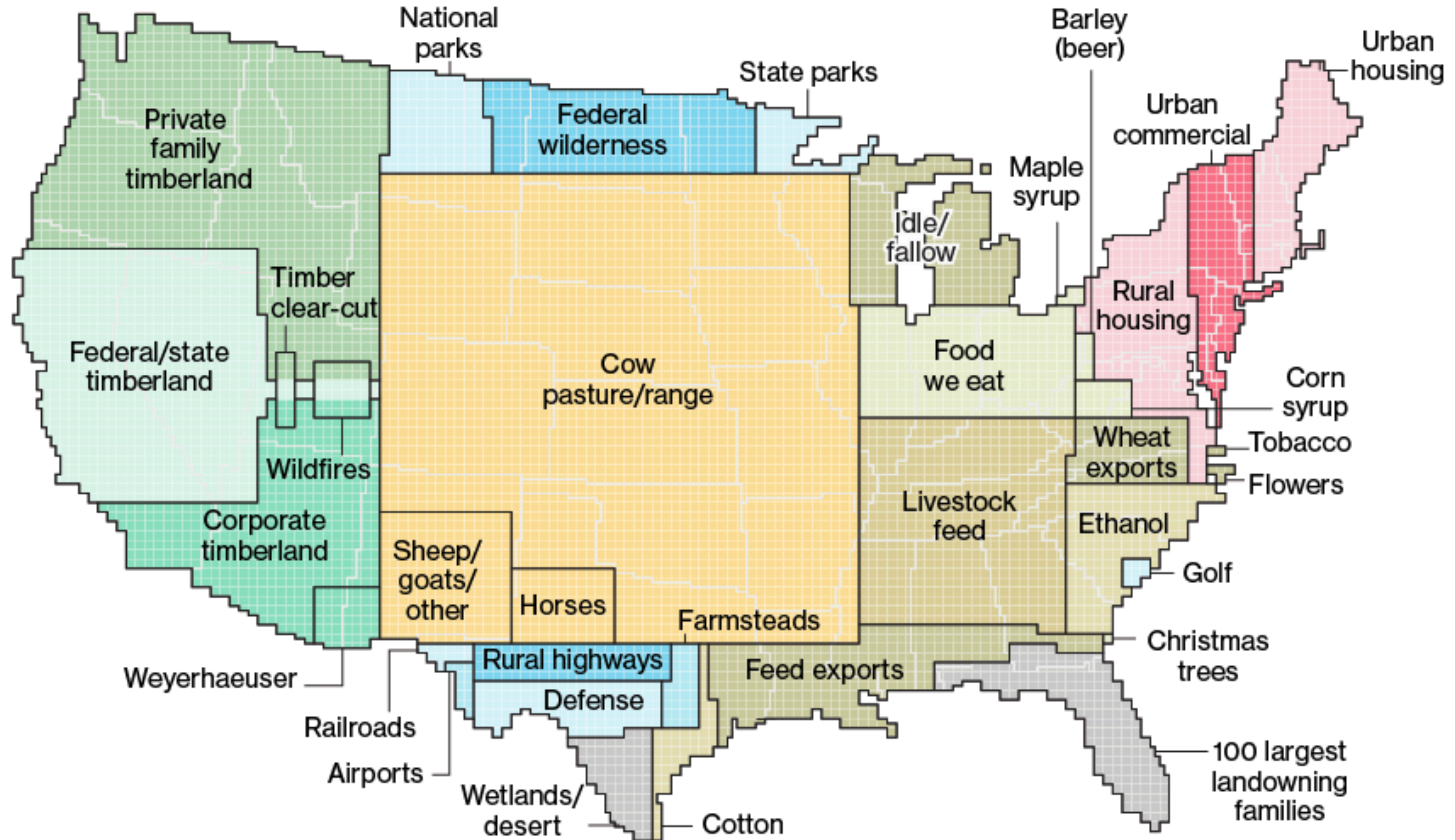
To explore each language branch, select a language family from the buttons above.



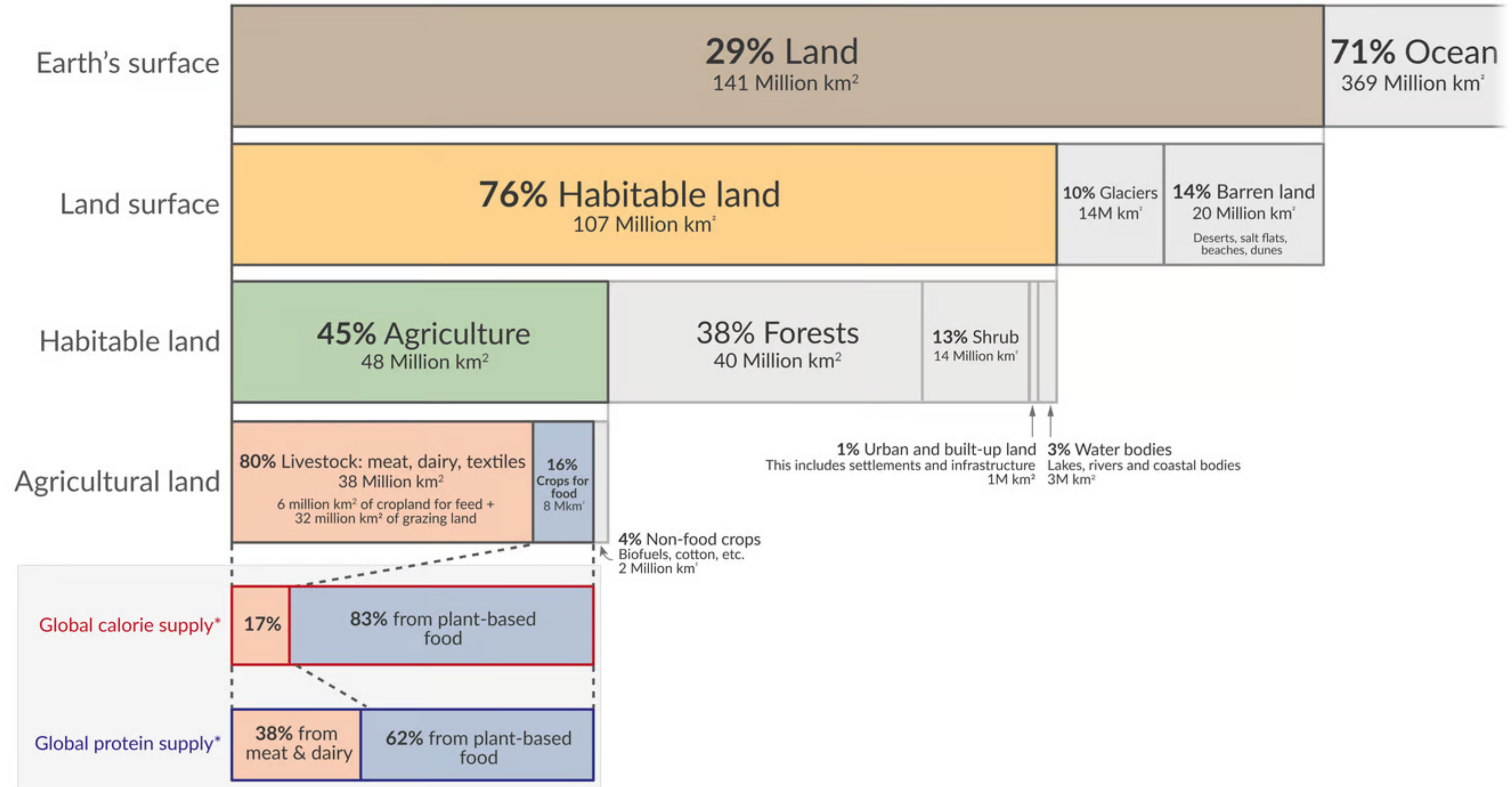
2024's solar eclipse



Here's How America Uses Its Land



Global land use for food production



*Includes fish and seafood from aquaculture production, which uses land for feed. If wild fish catch is also included, animal products would provide 18% of calories and 40% of protein.

Data sources: UN Food and Agriculture Organization (FAO) and Poore and Nemecek (2018).

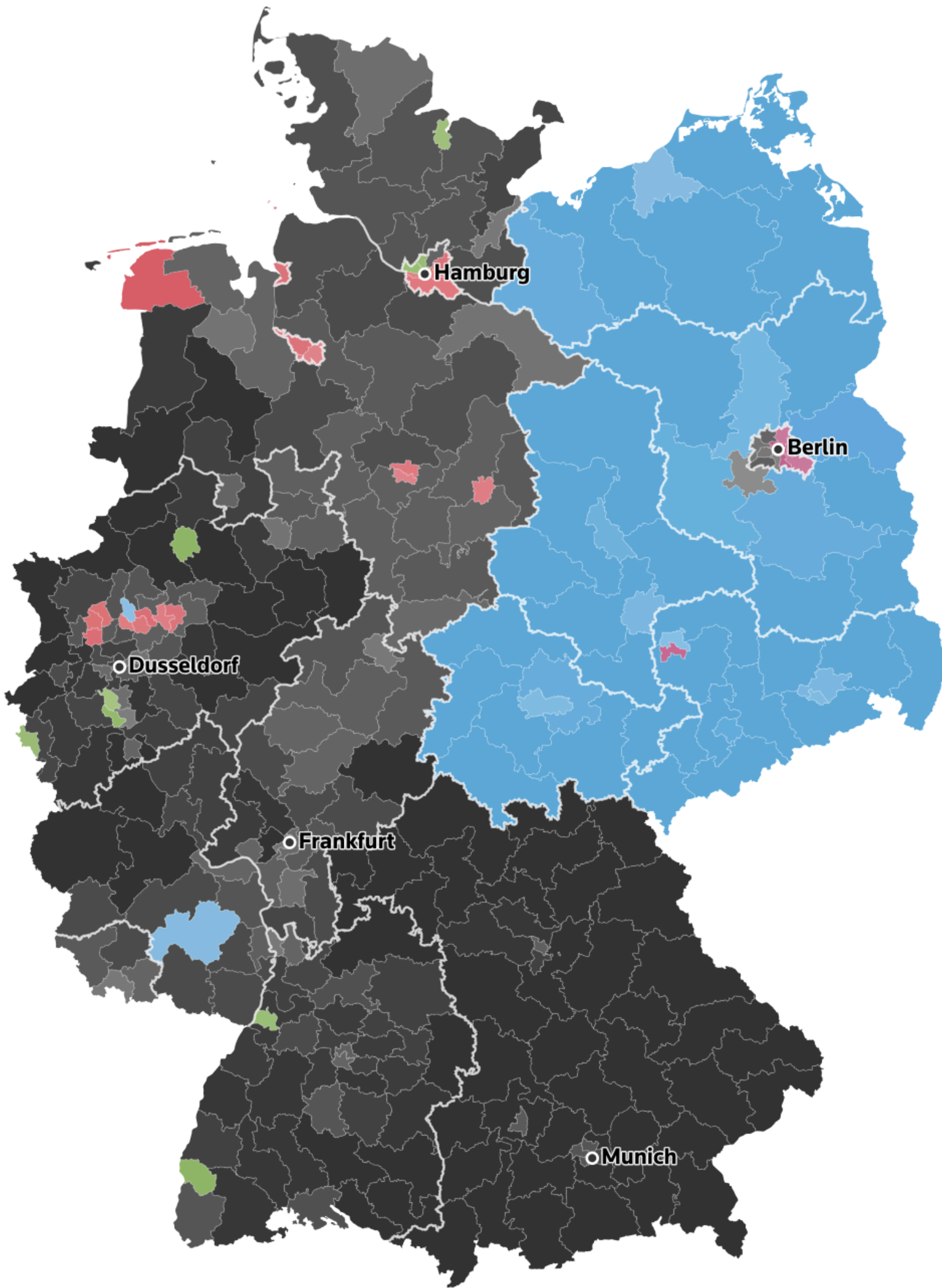
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German election map

National vote share

299/299 constituencies declared

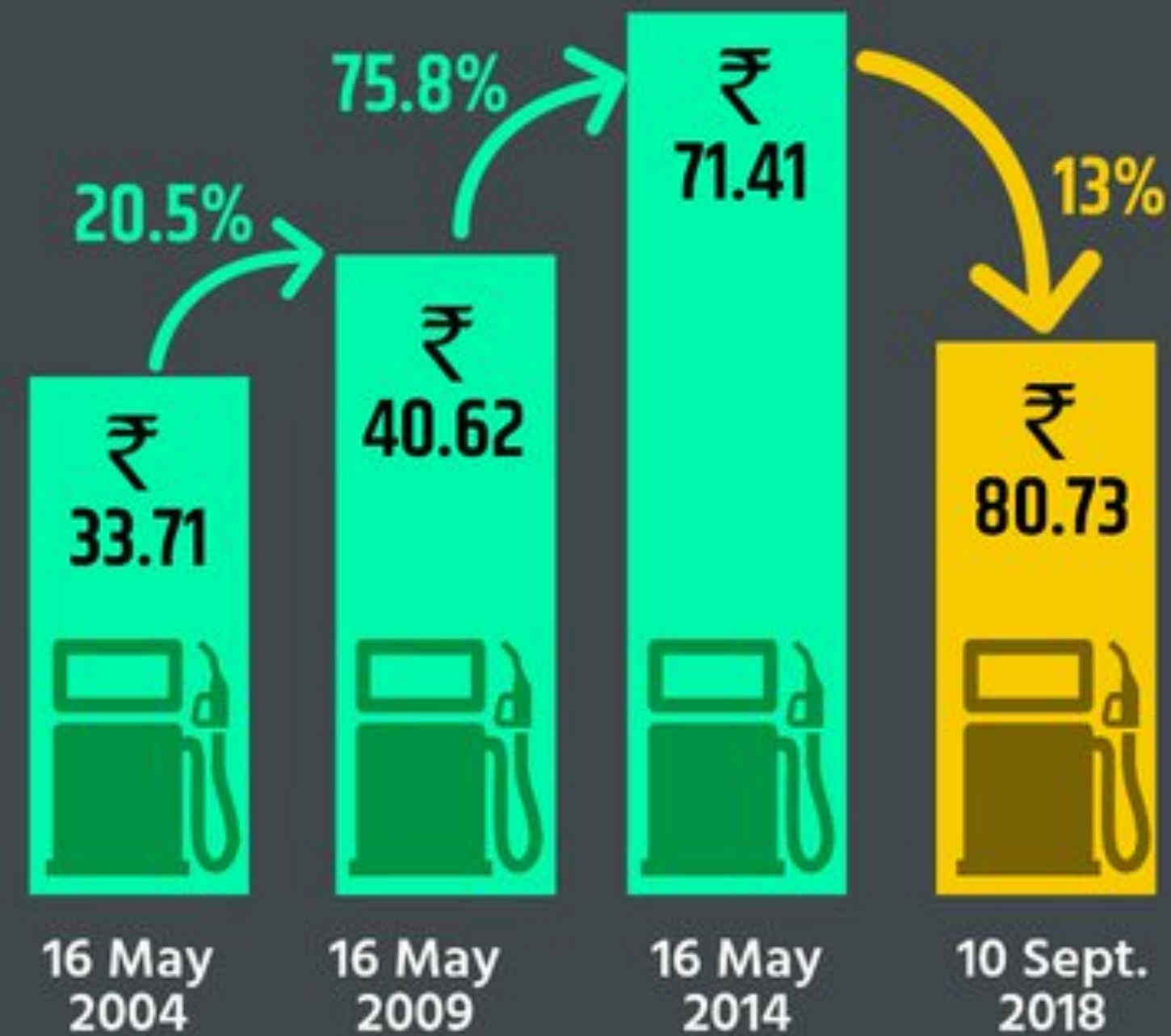


Really bad guys

Misleading visualizations

Truth of Hike in PETROLEUM PRICES

% Increase in Petrol Prices



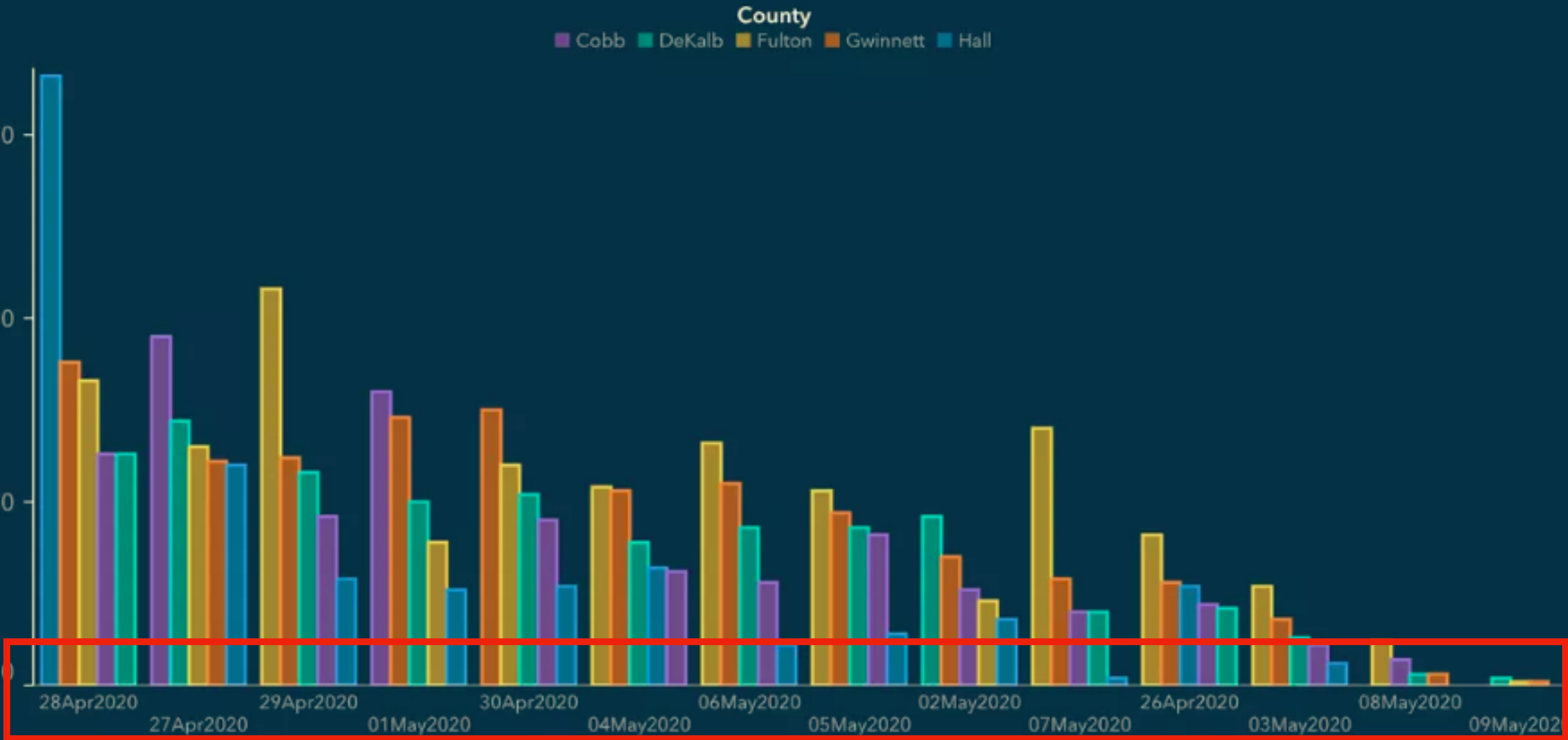
(Retail Selling Price in Delhi)

Drop??



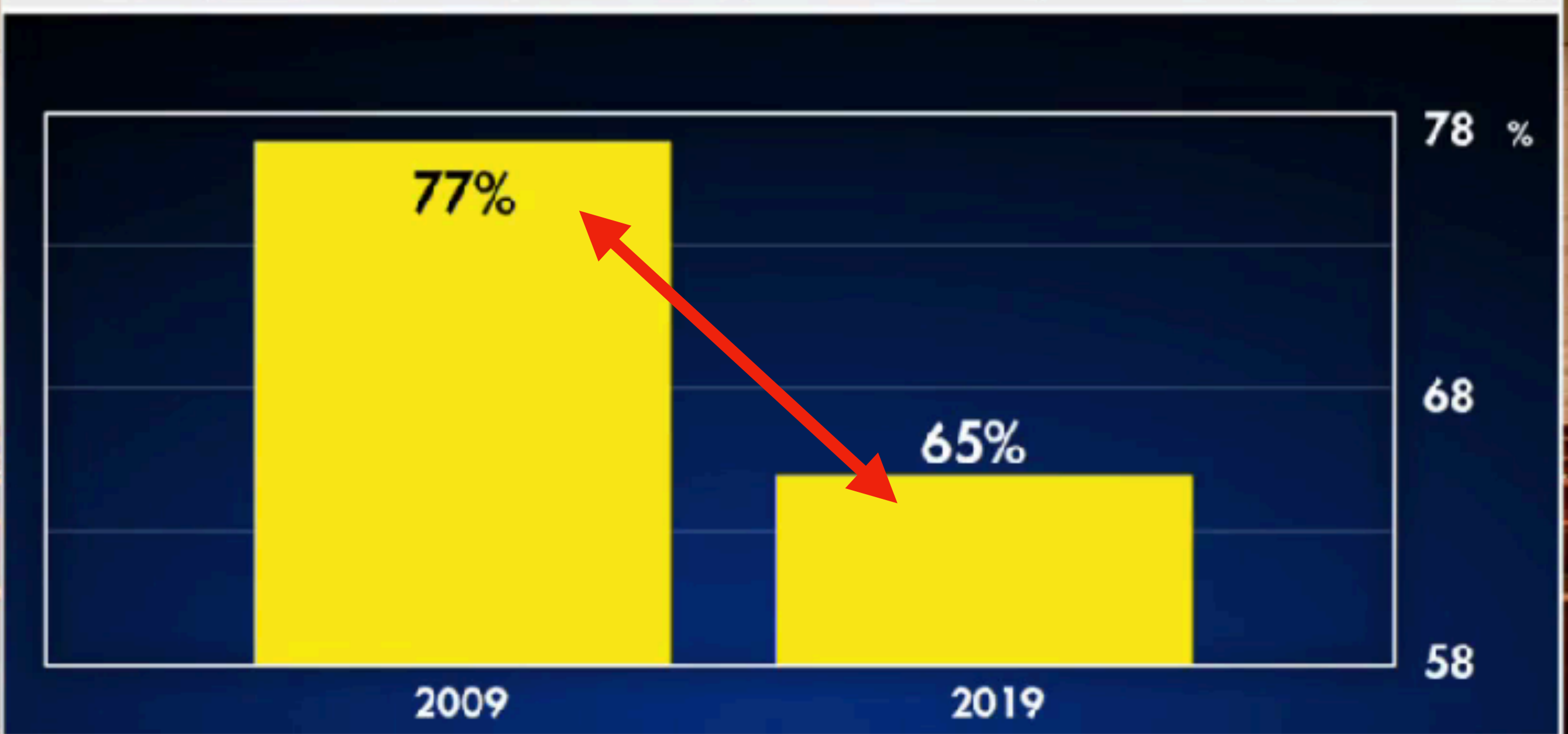
Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

The chart below represents the most impacted counties over the past 15 days and the number of cases over time. The table below also represents the number of deaths and hospitalizations in each of those impacted counties.





PERCENTAGE OF AMERICANS THAT IDENTIFY AS CHRISTIANS



PEW RESEARCH CENTER



WE USED TO BE AN ENTHUSIASTICALLY CHRISTIAN NATION

TUCKER CARLSON • TONIGHT •

Tucker Carlson: The cult of coronavirus has become its own religion

They have three samples

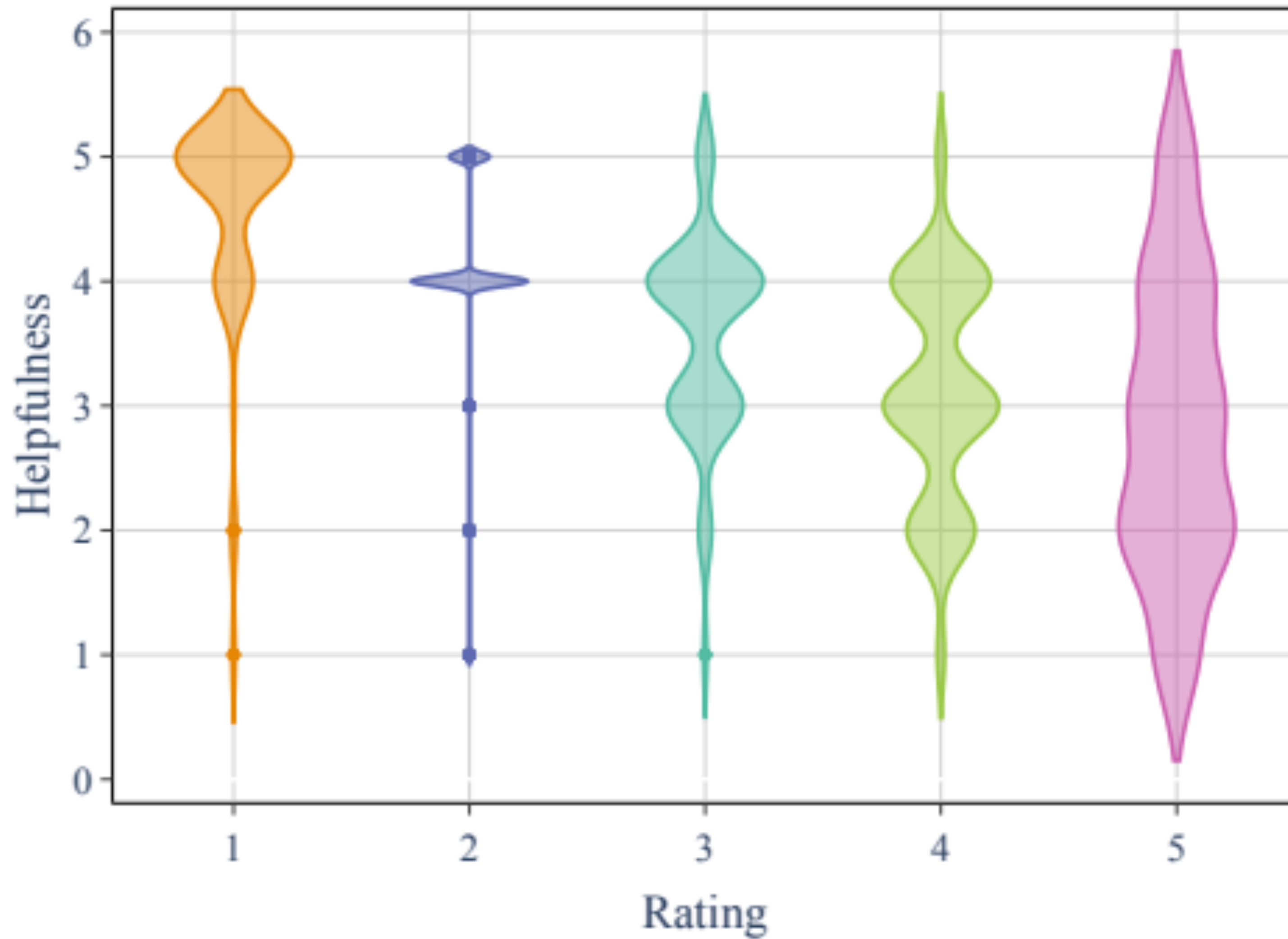


Tableau Lab

What is Tableau?

A powerful visual analytics and business intelligence (BI) tool

Helps users explore and understand **data interactively**

Supports **drag-and-drop** visualization design

Enables dashboard creation and data storytelling

No coding required, intuitive interface

Business analytics (sales trends, KPIs); **scientific research**; **education**

Setup

Faster / desktop: download

<https://terpware.umd.edu/search?q=Tableau%20Installer>

Slower / web-based: sign up

<https://public.tableau.com/app/discover>

